ACADEMIA RESEARCH PUBLISHING CENTRE

Journal of Interdisciplinary Research

Vol. 1, No. 1 (2025)
pp. 12–14 · ISSN 2949-4176 (online)

RESEARCH ARTICLE

Remote Work and Regional Labour Markets: Evidence from Online Vacancies

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Received: 21 October 2024 **Accepted:** 14 February 2025 **Published:** 20 March 2025

ABSTRACT

The spread of remote work after 2020 has the potential to reshape the spatial structure of labour markets by giving residents of smaller towns access to vacancies posted by employers in the capital. Using 2.4 million job advertisements published on a large online platform in 2019–2023, this article estimates the extent to which remote vacancies redistribute labour demand across regions. Advertisements were classified with a dictionary-based model followed by manual validation of a random sample. The share of fully remote vacancies rose from 2.1% in 2019 to 9.7% in 2023, with growth concentrated in information technology, marketing and financial services. A difference-in-differences design shows that in regions with a higher share of employment in these industries, the average offered wage grew 4.3% faster than in comparable regions. At the same time, remote vacancies impose higher requirements for experience and education, which limits access for a large proportion of workers. The equalising effect on interregional wage differences is statistically significant but moderate. The results suggest that remote work can serve as an instrument of regional policy only when combined with retraining programmes and the development of digital infrastructure.

Keywords: remote work; regional labour market; online vacancies; wages; difference-in-differences

Удалённая работа и региональные рынки труда: данные онлайн-вакансий

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АННОТАЦИЯ

Распространение удалённой работы после 2020 г. потенциально меняет пространственную структуру рынков труда, позволяя жителям небольших городов получать доступ к вакансиям столичных работодателей. В статье на основе 2,4 млн объявлений о вакансиях, опубликованных на крупной онлайн-платформе в 2019–2023 гг., оценивается, в какой мере удалённые вакансии перераспределяют спрос на труд между регионами. Для классификации объявлений использована словарная модель с последующей ручной проверкой случайной выборки. Доля полностью удалённых вакансий выросла с 2,1 % в 2019 г. до 9,7 % в 2023 г., причём рост был сосредоточен в сфере информационных технологий, маркетинга и финансовых услуг. С помощью метода разности разностей показано, что в регионах с более высокой долей занятых в этих отраслях средняя предлагаемая заработная плата выросла на 4,3 % сильнее, чем в сопоставимых регионах. Вместе с тем удалённые вакансии предъявляют более высокие требования к опыту и образованию, что ограничивает доступ к ним для значительной части работников. Эффект выравнивания межрегиональных различий в оплате труда оказался статистически значимым, но умеренным. Результаты свидетельствуют о том, что удалённая работа может стать инструментом региональной политики только в сочетании с программами переподготовки и развития цифровой инфраструктуры.

Ключевые слова: удалённая работа; региональный рынок труда; онлайн-вакансии; заработная плата; разность разностей

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For citation: Gromov P., Becker L. Remote Work and Regional Labour Markets: Evidence from Online Vacancies. *J. Interdiscip. Res.* 2025;1(1):12–14.

1 INTRODUCTION

The rapid shift to remote work during 2020 was initially regarded as a temporary response to the pandemic. Four years later, a sizeable share of employers continue to offer fully remote positions, raising the question of whether location still determines access to well-paid jobs. If workers in smaller towns can compete for vacancies posted in the capital, interregional wage gaps might narrow without physical migration.

Existing evidence on this question comes mainly from surveys of workers. Online vacancy data offer a complementary demand-side perspective with high temporal and spatial resolution. This article uses such data to document the growth of remote vacancies, to characterise their requirements and to estimate their effect on regional wage dynamics.

2 MATERIALS AND METHODS

The dataset comprises 2.4 million advertisements published between January 2019 and December 2023 on a large job platform. Each record includes the job title, description, region, industry, offered wage range and requirements. Remote vacancies were identified with a dictionary of 46 phrases; a manually coded random sample of 2,000 advertisements showed a precision of 0.94 and a recall of 0.91.

To estimate the effect on wages, regions were divided by their pre-pandemic share of employment in remote-intensive industries. A difference-in-differences model with region and month fixed effects compares the evolution of the median offered wage before and after March 2020, controlling for the occupational composition of vacancies.

3 RESULTS

The share of fully remote vacancies grew steadily from 2.1% in 2019 to 9.7% in 2023 (Fig. 1). Growth was strongest in information technology, where remote positions reached 31% of all postings, followed by marketing and finance. In manufacturing, retail and healthcare the share remained below 2%.

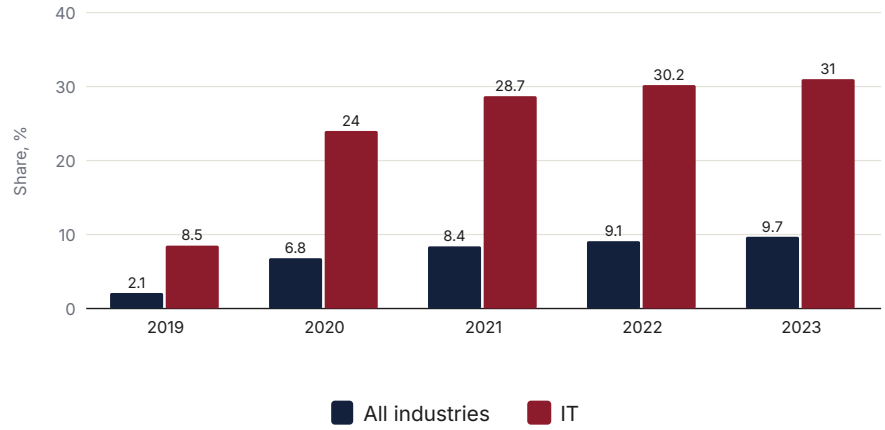


Fig. 1. Share of fully remote vacancies by year, %

Table 1 reports the main estimates. In regions with a high exposure to remote-intensive industries, the median offered wage increased by 4.3% more than in low-exposure regions. Remote vacancies were 1.6 times more likely to require higher education and on average requested 1.2 more years of experience.

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Table 1. Difference-in-differences estimates (dependent variable: log median offered wage)

Specification	Estimate	Std. error	N (region-months)
Baseline	0.043	0.012	4,980
+ occupation controls	0.037	0.011	4,980
Excluding capital region	0.041	0.013	4,920
Placebo (2018 break)	0.004	0.010	4,980

4 DISCUSSION

Our results indicate that remote work does exert an equalising pressure on regional wages, but the magnitude is moderate. The effect is concentrated in regions that already had a pool of workers in remote-intensive occupations, suggesting that remote work amplifies existing human capital rather than creating new opportunities.

The higher skill requirements of remote vacancies are a key constraint. Policy measures that combine digital infrastructure investment with retraining are more likely to translate the potential of remote work into broad-based gains. A limitation of the study is that vacancy data reflect offered rather than realised wages.

5 CONCLUSION

Remote vacancies have become a stable feature of the labour market and modestly reduce interregional wage gaps. Their benefits, however, accrue mainly to highly skilled workers, and their role as a regional development tool depends on complementary policies.

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